

Carlos Stolper, Dipseh Bhattarai, SooAh Lay, Jack Mclaud

Rocket Dashboard User Manual

1. Problem Statement

What it does:

The Rocket Dashboard is a desktop application that brings together all essential flight information for a model-rocket launch into a single interface. It displays real-time sensor readings (pressure, temperature, acceleration, gyroscope), lets you manage power and subsystem commands, plots the rocket's trajectory and altitude profile, runs a synchronized launch countdown with live weather data, and streams video from the launch pad.

Why it matters:

Ground-station operators and mission controllers must monitor multiple data streams simultaneously. Juggling separate tools increases the risk of missing a critical alert. This Dashboard unifies every measurement and control into one application, so you can focus on rapid decision-making, ensure a safe launch, and coordinate post-flight recovery without switching contexts.

2. Background & Motivation

In typical rocketry setups, you need separate instruments and software for sensor logs, maps, countdown clocks, and video feeds. Our team defined these user stories to guide development:

1. Power Management: As a ground-station operator, I want to turn avionics on and off with a single click.
2. Environmental Monitoring: As an engineer, I want live pressure and temperature gauges to verify rocket integrity pre-launch.
3. Attitude Tracking: As a flight director, I want continuous accelerometer and gyroscope plots to watch rocket orientation.
4. Trajectory Mapping: As a mission planner, I want a world-map plot of latitude/longitude so I can predict landing zones.
5. Countdown & Weather: As a launch director, I want a synchronized countdown timer with local weather and sunset time.

6. Live Video: As a broadcast engineer, I want a video stream of the launch to share with spectators.
7. Recovery Assistance: As a recovery team member, I want the last-seen GPS coordinates to locate the rocket after landing.

Use-Case Summary:

- Operator interacts with the Dashboard to power systems, start/stop recordings, reset countdowns, and view data.
- System provides sensor readings, GPS fixes, weather data, countdown timing, and video frames in response.

3. Installation & Getting Started

Prerequisites

- Java 11 or newer installed and on your PATH
- Git client (or GitHub Desktop)
- Internet access for weather API calls

Clone & Build

bash

```
git clone https://your-repo-url/csci205_final_project.git
cd csci205_final_project
./gradlew clean build
```

Launch the Application

bash

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```
./gradlew run
```

Alternatively, in IntelliJ IDEA:

Open the project folder.

Let Gradle sync dependencies.

Right-click on AirForce.view.RocketDashboardApp → Run.

Resource Locations

src/main/resources/Avionics.csv — raw sensor log (13 columns per row)

src/main/resources/videos/LiveFeed.MOV — launch-site video file

src/main/resources/Avakas.png — application logo

To add new data, place your CSV or video files here; the Dashboard loads them automatically by filename.

4. Using the Dashboard

When the Dashboard window opens, you'll see seven main panels:

4.1 System Operation

Voltage Dial (0–12 V): shows current bus voltage.

Power ON / OFF Buttons: click “Power ON” to initialize all subsystems; click “Power OFF” to safely shut down.

4.2 Command Operation

Four toggles control individual modules:

BM P180 (Bus Manager)

MPU 6050 (Motion Processing Unit)

TLM 500 (Telemetry Transceiver)

GPS (Global Positioning System)

Each button switches between ON and OFF, updating its label accordingly. Use these to arm or disarm hardware modules during setup.

4.3 Sensor Logs

Four real-time displays:

Pressure Gauge: analog dial showing pressure in kilopascals (kPa).

Temperature Gauge: semicircular meter showing temperature in °C.

Gyro Scope Plot: line graph of GyroX, GyroY, GyroZ over time.

Accelerometer Plot: line graph of AccX, AccY, AccZ over time.

Under each plot, click Record to start or stop live data capture. Numeric readouts below gauges update continuously when recording.

4.4 Rocket Trajectory & Launch Graph

World Map Plot: shows the rocket's latitude and longitude path as recorded points.

Coordinates & Speed: text fields display the latest longitude, latitude, velocity (km/h), and altitude (km).

Trajectory Record Button: click "Record" to append new GPS fixes to the map.

Launch Graph Panel: plots altitude vs. time once data is available.

4.5 Countdown Clock & Live Feed

Countdown Display: large digital timer in HH:MM:SS counting down to launch.

Weather Summary: current temperature, forecast icons, and time until sunset.

Live Video Pane: streams your LiveFeed.MOV file.

When T-20 s is reached, video auto-plays.

Press Power OFF or reset the countdown to stop playback.

4.6 Timeline

Vertical list of pre-launch events (e.g., "1 pm setup", "2 pm setup", "3 pm setup"). Use this for quick status reference during checks.

4.7 Recovery

Launch Coordinates: the static lat/long where the rocket took off.

Last Seen Coordinates: the most recent GPS fix, updated when you record trajectory.

Use these to guide the recovery team to the rocket's landing site.

5. Tips & Troubleshooting

Empty Gauge Readings?

Confirm Avionics.csv is present under resources/.

Check console for parsing errors (invalid number of columns).

Countdown Doesn't Start or Reset

Click Power ON first.

Use the Power OFF button to stop, then Power ON again to restart.

Video Playback Errors

Ensure LiveFeed.MOV is located in resources/videos.

If unsupported, convert to MP4/H.264:

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```
ffmpeg -i LiveFeed.MOV -vcodec libx264 -acodec aac LiveFeed.mp4
```

Map Remains Blank

Toggle GPS ON.

Hit the Record button under the map panel to begin capturing fixes.

Application Fails to Launch

Run with stacktrace:

bash

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```
./gradlew clean run --stacktrace
```

Verify you have JavaFX modules on your PATH: javafx.controls, javafx.fxml, javafx.media.